

Practical 3_2

```
[3]: import pandas as pd  
d = pd.read_csv("pract3.csv") d
```

```
[3]:
```

	age	income	gender
0	53	75354	male
1	40	57514	female
2	54	74673	male
3	46	64249	male
4	69	64249	male
5	69	73370	female
6	63	75187	female
7	60	50441	male
8	52	67180	female
9	35	64548	male
10	63	79660	female
11	54	60999	male
12	54	65535	male
13	67	76537	female
14	66	53068	male
15	69	55539	male
16	30	66413	female
17	64	56067	female
18	62	68205	male

```
[4]: d['income'].mean()
```

```
[4]: 65725.68421052632
```

```
[5]: d['age'].mean()
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```
[5]: 56.31578947368421
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[6]: d['income'].median()
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[6]: 65535.0
```

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[7]: d['age'].median()
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[7]: 60.0

[8]: d['income'].min()

[8]: 50441

[9]: d['age'].max()

[9]: 69

[10]: d['income'].max()

[10]: 79660

[11]: d['income'].sum()

[11]: 1248788

[12]: d['age'].max()

[12]: 69

[13]: d['age'].sum()

[13]: 1070

[14]: d['age'].std()

[14]: 11.666917290541125

[15]: d['income'].std()

[15]: 8586.003423224669

[16]: d.groupby(['gender'])['age'].mean()

[16]: gender
      female    56.000000
      male     56.545455
      Name: age, dtype: float64

[17]: from sklearn.preprocessing import LabelEncoder
      le = LabelEncoder()
      d['gender']=le.fit_transform(d['gender'])
      d

[17]:   age  income  gender
0   53   75354        1
1   40   57514        0

```

2	54	74673	1
3	46	64249	1
4	69	64249	1
5	69	73370	0
6	63	75187	0
7	60	50441	1
8	52	67180	0
9	35	64548	1
10	63	79660	0
11	54	60999	1
12	54	65535	1
13	67	76537	0
14	66	53068	1
15	69	55539	1
16	30	66413	0
17	64	56067	0
18	62	68205	1

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